

Errata to “The Universality of Thomas L. Etter’s Three-Place Identity: A Formal Proof and Its Significance”

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These corrections concern the appendices and the accompanying Lean 4 modules of the above paper. They fall *entirely* on the present author’s reconstructions and formalization; *none* of them touches Etter’s own constructions, which they leave intact and, in several places, restore to their full strength. Because a list of corrections read in isolation can do an injustice to the work it corrects, the standing record is stated first.

1 What Etter established, and what stands

Etter’s thesis — that *identity*, taken as a relative, three-place predicate, is expressively adequate for the whole of mathematics — is correct, and is of first-rank foundational significance. It inverts the usual order of foundations: not “things exist, then we ask whether they are identical,” but identity (from a viewpoint) as primitive, with membership, and hence set theory, *derived*. The following are Etter’s, are correct, and are unaffected by anything below.

- **The Universality Theorem (main body).** Definitions D1 and D2, Theorems 4.1–4.3, and the round-trip of §5 are a faithful transcription of Etter’s construction in *Three-place Identity* [E3]: membership is definable from three-place identity and *vice versa*, with the two round-tripping. The mechanization discharges this with a clean axiom profile. This is the paper’s central result and it is correct.
- **The three-equality expressiveness theorem ($\mathbf{RCV} \rightarrow \mathbf{ZF}$).** That ZF set theory is constructively definable from three equality predicates is *proved in full* by Etter in *The Expressive Power of Equality* [E1], via the cell construction. This is a complete theorem, not a conjecture. Its mechanization (`Cell.lean`) discharges Etter’s Theorems 1 and 2 with a clean `#print axioms`.
- **The two-equality Stereo Equality theorem.** A genuine research frontier, not an error. Etter announces it in *Three-place Identity* [E3, §3], defers its proof to Section 3 of *Membership and Identity* [E2], and sketches the key *link* construction in [E3, fn. 5]; the M&I draft was never completed (so noted therein by R. Shoup). It is stated precisely, and recorded as an open problem, below.
- **“Universality” as relative interpretability.** The paper’s use of “universal”/“open system” is Etter’s own rigorous notion of relative interpretability, and is the correct claim — expressive adequacy, not model-theoretic foundationalism.

None of the corrections that follow qualifies any of these.

2 What the corrections change

Where the paper went wrong was in the present author’s *secondary presentation* of Etter’s two- and three-equality constructions, in Appendices A and B and their Lean modules. There the author substituted a *symmetric* intersection formula — a reconstruction of his own, which does not appear in Etter’s papers — for Etter’s actual, *directed* constructions, and a fragmentary mechanization for a complete one. The corrections below remove that authorial error and *restore* Etter’s stronger original work.

Erratum 1 — the author’s intersection membership is not a ZF membership

As published. Appendix A defines $y \in' x \iff \neg(E_1(y, x) \wedge E_2(y, x))$, and §B.3 defines $y \in_{\text{RCV}} x \iff \neg(R(y, x) \wedge C(y, x))$, each asserted to satisfy the axioms of ZF.

Correction. These formulas were introduced by the present author. Because E_1, E_2 (resp. R, C) are equivalence relations, hence symmetric, each defined relation is symmetric, $y \in' x \iff x \in' y$. ZF membership is not symmetric ($\emptyset \in \{\emptyset\}$ but $\{\emptyset\} \notin \emptyset$), so these relations cannot be ZF memberships, and the assertions that they “satisfy the axioms of ZF” are withdrawn. The defect is one of the author’s reconstruction; Etter’s own constructions (Erratum 2) are directed, and are unaffected.

Erratum 2 — Etter’s actual RCV $\rightarrow$ ZF construction (restored)

Correction. The intersection formulas above do not occur in Etter’s surviving papers. Etter’s RCV $\rightarrow$ ZF membership is the *cell construction* of *The Expressive Power of Equality* [E1]: a cell is an ordered triple $\langle x, y, n \rangle$ with $x \varepsilon y$ and $n \in \{1, 2\}$, of value $\text{Val}(\langle x, y, n \rangle) = x$ if $n = 1$ else y ; R, C, V are the row, column and value equalities on cells, with V taken as identity; and membership is recovered from R, C, V alone by

$$\begin{aligned} \text{First}(c) &\iff \forall d \exists e (V(e=d) \wedge C(e=c)), \\ c E d &\iff \exists e, f (R(e=f) \wedge \text{First}(e) \wedge V(c=e) \wedge \neg \text{First}(f) \wedge V(d=f)). \end{aligned}$$

The argument-asymmetry supplied by First/ $\neg$ First is exactly the directedness the author’s symmetric formula lacked. Appendix B should cite [E1] and use this construction; the corrected appendix does.

Erratum 3 — status of the appendix results

Correction. The three-equality result — that ZF is constructively definable in an RCV language — is not merely asserted: it is proved in full in [E1] by the cell construction. The only genuinely open item is the *two-equality* Stereo Equality theorem. Its construction is the *link sketch* of [E3, fn. 5]: an equality *without transitivity* $[x=y]$ (reflexive and symmetric only), a “meet” operator $[x=:y, z, \dots]$ (“ x equals each of $y, z, \dots$ and anything unequal to y is unequal to x ”), and a link membership

$$x \in' y := \exists x', x'', y', y'', q, q'. [q=:x, q'] \ \& \ [q'=:y, q, q''] \ \& \ [x'=:x, x''] \ \& \ [x''=:x, x'] \ \& \ [y'=:y, y''] \ \& \ [y''=:y, y'],$$

with the claim that there is a set model of $[x=y]$ in which $x \in' y$ is ZF membership. The surviving M&I draft breaks off before proving this, so the honest status is an *open conjecture at the frontier* — not an error, and not a claim the paper was entitled to assert as proved.

Erratum 4 — the Lean modules

Correction. As published, `Stereo.lean` and `RCV.lean` formalized the author’s symmetric intersection formula and proved only disconnected fragments (a trivial irreflexivity, a definitional link lemma, partial uniqueness); they neither related membership to the link, nor established any ZF axiom, nor stated the symmetry that refutes the formula, nor contained Etter’s cell apparatus. The main-body round-trip module is unaffected and stands. A faithful mechanization now accompanies the repository [E6]: `Cell.lean` discharges Etter’s Theorems 1 and 2 (the cell construction) over Mathlib’s `ZFSet`; `Stereo.intersectionMem_not_ZF` records that the author’s symmetric intersection candidate cannot be a ZF membership; and the two-equality result is stated as a precise, falsifiable proposition (`etterStereoConjecture`) transcribing the [E3, fn. 5] link construction.

Net effect. These corrections *raise* the standing of the program. Where the paper claimed a result, Etter’s own work already supplies a complete proof [E1]; where the paper reached past what is proved, the honest status is an open conjecture at the frontier of a still-living research program. The errors were the present author’s, in a secondary presentation; Etter’s constructions stand, and are stronger than the reconstruction that briefly stood in for them.

References

- [E1] Etter T. *The Expressive Power of Equality*. Boundary Institute; June 20, 2001. (Contains the cell construction and the proof of the expressiveness/RCV theorem.)
- [E2] Etter T. *Membership and Identity*. Boundary Institute; 2006. (Incomplete draft.)
- [E3] Etter T. *Three-place Identity (Set Theory and the Identity of Mathematical Objects)*. Boundary Institute; 2006.
- [E4] Etter T. *Notes on Identity and Pairing*. Boundary Institute; 2001.
- [E5] Etter T. *Equalities and Quine Identities*, §§1–6. Boundary Institute; 2001.
- [E6] Bowery J. *RelativeIdentity* (Lean 4 sources; `Cell.lean`, `RCV.lean`, `Stereo.lean`). <https://github.com/jabowery/RelativeIdentity>